

SESSION XI: Analog Circuit Techniques

THAM 11.5: A High-Accuracy Analog Multiplier

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MANY ANALOG MULTIPLIERS currently available use the *linearized transconductance* principle first discussed earlier at ISSCC¹. It was then generally believed to be limited to low-to-medium accuracy uses. This paper will describe a monolithic four-quadrant multiplier based on this principle having accuracy specifications usually associated with the classical high-precision techniques such as pulse-width-height multiplication. Although similar to most transconductance multipliers, the performance is achieved through a combination of accurate support circuitry, careful layout design and attention to detail. An active feedback scheme reduces nonlinear errors to negligible proportions.

The widely-used *inverted* multiplier core (Figure 1) was retained; it allows the use of simple, wideband all-NPN input amplifiers. The major sources of error in this circuit have already been described¹. Emitter-area mismatch causes input offsets and a parabolic nonlinearity. To reduce the latter to $\pm 0.05\%$ FS requires that area-ratios match to $\pm 0.1\%$, equivalent to a V_{BE} mismatch of $25 \mu V$. These transistors therefore have large emitters, photo-repeated at artwork level to eliminate cutting errors, and are arranged in two overlapping quads to reduce the effects of processing gradients². Ohmic emitter-resistance mismatch introduces an odd-harmonic nonlinearity, which can be minimized by operating the X and Y input amplifiers at identical current, using a wrap-around base contact and operating at low current-densities. Metalization resistances are balanced to $\pm 20 m\Omega$ and four-terminal connections to Q_1 and Q_2 are used.

The thermal integrity of the core is of utmost importance. For example, an input offset of 0.1% FS will be introduced by only $0.025^\circ C$ between Q_1 and Q_2 . This, and other thermal errors, are kept within acceptable bounds by the use of the cross-connected quads, which are placed at the center of a thermally symmetric layout, and the quiescent power dissipation is almost exactly equal in each of the surrounding quadrants.

The basic circuit suffers from beta-sensitivity: the gain is reduced by the factor $\beta/(3 + \beta)$, which typically introduces a temperature coefficient of $+0.012\%/^\circ C$. Compensation is introduced to achieve $0-70^\circ C$ gain stabilities of the order of $\pm 0.2\%$. Overall scale-factor is determined by an internally-generated

reference voltage using a special high-stability breakdown diode. Its supporting circuitry is shown in Figure 2. This serves to provide temperature compensation for the reference diode, insure that this diode operates at a supply-insensitive current, and introduce some of the needed beta-compensation.

Emitter-degenerated amplifiers (Figure 3) are used to convert input voltages into a pair of complementary currents. However, they introduce distortion as the signal approaches the limit of the dynamic range, $IR/2$, due to the nonlinear base-emitter voltages:

$$V = (x - \frac{1}{2})IR + \frac{KT}{q} \ln \frac{x}{1-x}$$

Figure 4 shows that not all of the second term represents non-linearity. The nonlinear contribution is:

$$\Delta V = \frac{KT}{q} \left(\ln \frac{x}{1-x} - 4x + 2 \right)$$

To achieve an error of 0.02% FS ($\pm 10V$), ΔV can be only $2 mV$. This translates to $x = 0.5 \pm 0.235$ at FS, so the dynamic range is over twice the needed $10V$ and offset, drift and noise are doubled. The core is capable of linear operation right up to $x = 0.5 \pm 0.5$, so a solution to this problem is needed. More complex input stages could be used, but these would probably have inferior dynamics. The approach adopted solves the linearity problem without changing the input stages, which are now given a dynamic range of only slightly more than $10V$.

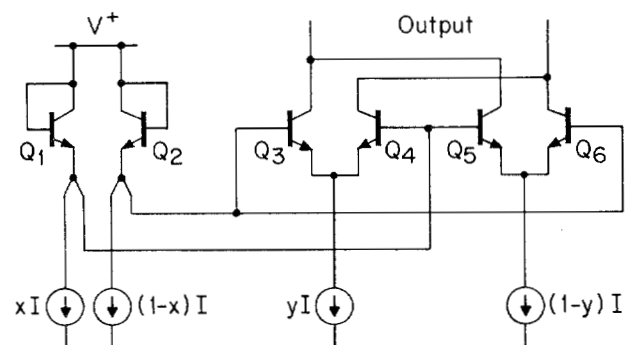


FIGURE 1—Standard transconductance multiplier core.

¹ Gilbert, B., "A Precise Four-Quadrant Multiplier with Subnanosecond Response", *Journal of Solid State Circuits*, p. 365-373; Dec., 1968.

² Maidique, M. A., "A High-Precision Monolithic Super-Beta Operational Amplifier" *Journal of Solid State Circuits*, p. 480-487; Dec., 1972.

³ Analog Devices Type AD534.

Figure 5 shows a simplified schematic of the complete multiplier³. The feedback round the output op-amp is via the extra amplifier identical to the X and Y input amplifiers. It therefore has the same nonlinearities. It will be shown that the overall error is zero when either the X or Y input is at $\pm 10V$, and never exceeds 0.05% FS. This approach has several other advantages. The sensitivity of output offset voltage to supply voltage is very low; the scale-factor can be readily changed (e.g.,

to unity) by an external feedback attenuator; the unused feedback terminal permits the addition of a signal to the output; as a feedback divider, numerator and denominator are both fully differential and at high impedance; the transient response is improved due to the cancellation of equal zeroes in the emitter circuits of the input and feedback stages.

The output op-amp is a complete high-gain medium-bandwidth circuit with full output protection. Ideally, an order of magnitude improvement in static accuracy ought to be accompanied by a similar improvement in dynamic accuracy; e.g., settling time. However, static accuracy will be in jeopardy if the dissipation has to be greatly increased to achieve wide bandwidth, and chip space is limited. A compromise design was used, giving an open-loop gain (from the feedback input) of 10,000 and an f_T of about 2 MHz. This is adequate for most applications and insures accurate divider performance over a 100:1 denominator range.

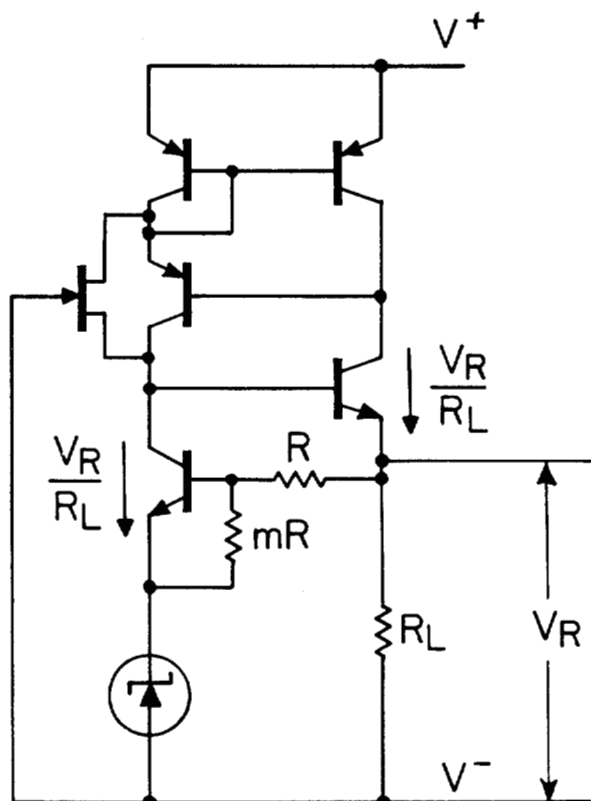


FIGURE 2—Reference generator.

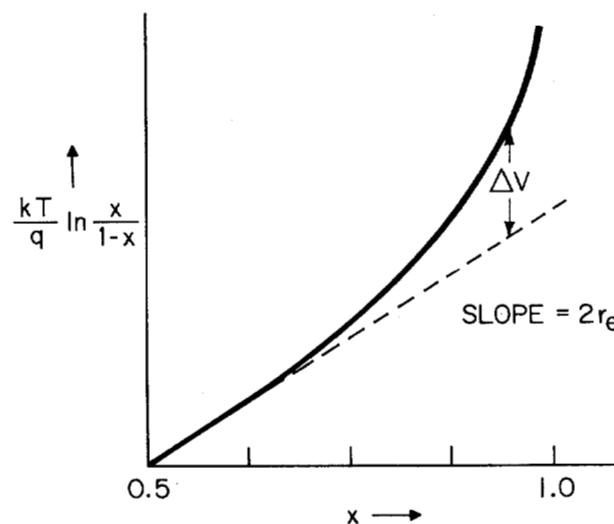
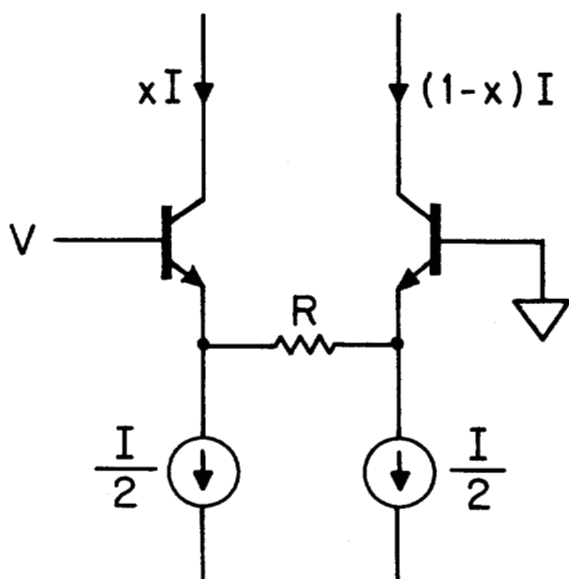
FIGURE 4—Distortion due to r_e .

FIGURE 3—Emitter degenerated amplifier.

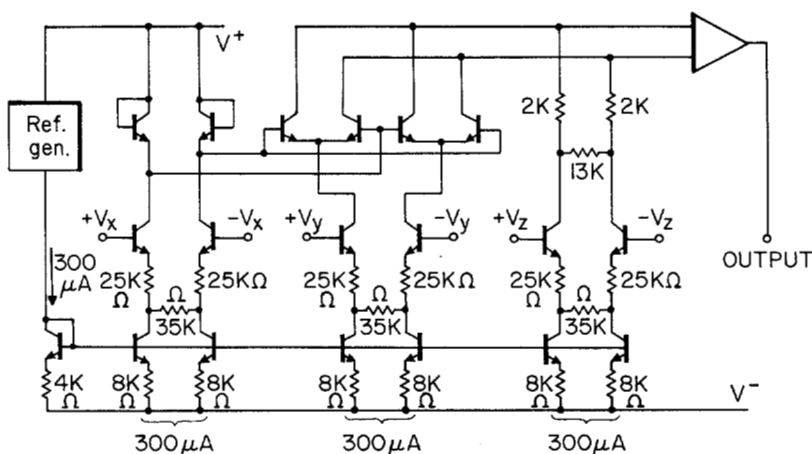


FIGURE 5—Simplified multiplier schematic.